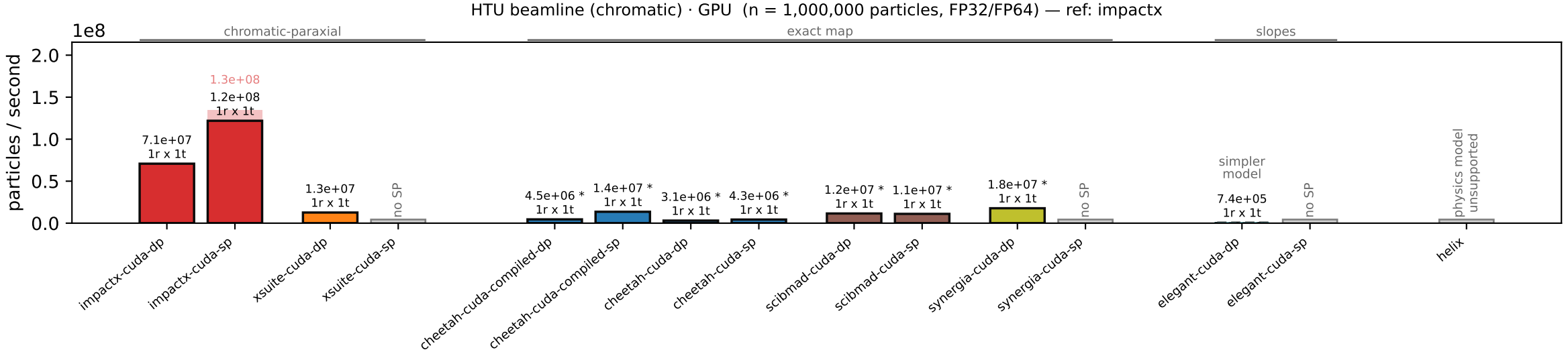


*'simpler model' = a lower-fidelity model vs the reference — elegant: tracks geometric slope (x,x') not canonical (x,px)*

*lighter bar behind + lighter value above = fast-math (relaxed FP), shown when >=5% faster than the IEEE bar*

** Cheetah/SciBmad/Bmad/Synergia have no chromatic-paraxial model here; they run the costlier exact map (agrees within tol)*

versions: impactx 26.08 · xsuite 0.53.4 · cheetah 0.8.2 · scibmad 0.4.1 · synergia v2024.04-209-g795b99f39 · elegant elegant-2026.4.0 · helix v1.10.1